Midterm Exam — Spring 2026

Q1. If A is an $n \times n$ singular matrix, then the linear system $A\underline{X} = 3\underline{X}$

(a) is homogeneous
(b) has a unique solution
(c) is inconsistent
(d) is nonhomogeneous
(e) None

Q2. A, B, and C are $n \times n$ nonsingular matrices. A is diagonal, B is lower triangular, and C is upper triangular. $(B^TC)^{-1}A$ is:

(a) lower triangular
(b) diagonal
(c) symmetric
(d) upper triangular
(e) None

Q3. If A is an $n \times n$ skew-symmetric nonsingular matrix, then $(A^{-1})^3$ is:

(a) triangular (b) skew-symmetric (c) diagonal (d) symmetric (e) None

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Q4. If $A = \begin{bmatrix} 2 & 1 & -7 \\ -1 & 3 & r \end{bmatrix}$ where $r \in \mathbb{R}$, then the entry in row 2 and column 3 of A^TA is equal to:

(a) $7 - 3r$ (b) $14 + r$ (c) $-14 - r$ (d) $-7 + 3r$ (e) None

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Q5. The linear system

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix} \underline{X} = \begin{bmatrix} 4 \\ 0 \\ 3 \\ 1w \end{bmatrix}$$

will be inconsistent if w is not equal to:

(a) -9 (b) -3 (c) -6 (d) 3 (e) None

Q6. If $(A^{-1} + 2I)^T = \begin{bmatrix} 1 & 4 \\ 0 & g \end{bmatrix}$, $g \neq 2$, then the entry in row 2 and column 1 of A is equal to:

(a) $4/(g-2)$ (b) $-4/(g-2)$ (c) $1/(g-2)$ (d) $-1/(g-2)$ (e) None

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Q7. Consider the linear system:

$$x + y - z = a$$
$$2x + 5y + 2z = -2$$
$$3x + 2y + z = 2$$

If $x = r$, $y = s$, and $z = 3$ is the solution of this system, where $r, s, a \in \mathbb{R}$, then a is equal to:

(a) 1 (b) -1 (c) -4 (d) 4 (e) None

Q8. If A and B are $n \times n$ matrices, then $(A^2 - B^2) = (A + B)(A - B)$

(a) True (b) False (c) (d) (e) None

Q9. Consider the linear system:

$$\begin{aligned} x + 2y - z - 2w &= 3 \\ 2x + 3y - 5z - 6w &= 5 \\ 3x + 4y - 9z - 10w &= 7 \end{aligned}$$

If $z = t$, $w = s$, and $s, t \in \mathbb{R}$, then x is equal to:

(a) $1 + 6s + 7t$ (b) $1 + s - 5t$ (c) $1 - s - t$ (d) $1 - s - 5t$ (e) None

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Q10. The linear system:

$$\begin{aligned} x - 4y + 2z &= 5 \\ x + 3y + 4z &= 6 \\ rx + sy + tz &= 7 \end{aligned}$$

will not have a unique solution if:

(a) $18r + 4s - 7t = 0$ (b) $22r + 2s - 7t = 0$

(c) $r - 2s + 3t = 0$ (d) $2r + 5s + 2t = 0$

(e) None

Q11. If the matrix $\begin{bmatrix} 1 & 3 & f^2 - 4 & 2 & 0 & 4 \\ 0 & 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & f + 2 & 2 \end{bmatrix}$ is in row echelon form (REF), then f is equal to:

(a) 2 (b) -3 (c) -2 (d) -1 (e) None

Q12. If $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$, the entry in row 2 and column 3 of E^{-1} is:

(a) $1/3$ (b) $-1/3$ (c) -3 (d) 3 (e) None

Q13. If A, B, C, and D are $n \times n$ nonsingular matrices, and $AXB + C = D$, then X is equal to:

(a) $A^{-1}(D - C)B^{-1}$ (b) $B^{-1}(D - C)A^{-1}$

(c) $A^{-1}(D + C)B^{-1}$ (d) $B^{-1}(D + C)A^{-1}$

(e) None

Q14. The product of an $n \times n$ symmetric matrix S and an $n \times n$ skew-symmetric matrix K is:

(a) symmetric (b) skew-symmetric (c) diagonal (d) a zero matrix (e) None

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Q15. Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 2 & 4 \\ 3 & 2 & 6 \\ 4 & 5 & f \end{bmatrix}$, where $|f| > 12$. The entry in row 3 and column 3 of L is equal to:

(a) $f - 9$ (b) $f - 8$ (c) $1/(f - 8)$ (d) $1/(f - 9)$ (e) None

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Q16. The linear system:

$$\begin{aligned} x - 2y + 4z &= a \\ 2x - 3y + 11z &= b \\ 3x - 9y + 3z &= c \end{aligned}$$

will be consistent if:

(a) $c = -9a + 3b$ (b) $c = -7a + 2b$ (c) $c = 7a - 2b$ (d) $c = 9a - 3b$ (e) None

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Q17. The linear system:

$$\begin{aligned} x + y + z &= 4 \\ x + 2y + z &= 3 \\ x + y + kz &= b \end{aligned}$$

will have no solution when:

(a) $k = 1$ and $b = 4$ (b) $k = 3$ and $b = 2$ (c) $k = 1$ and $b = 3$ (d) $k = 2$ and $b = 3$ (e) None

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Q18. Consider the LU decomposition of the matrix $\begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & 3 & 2 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 3 & -3 \end{bmatrix}$. The entry in row 3 and column 4 of U is equal to:

(a) -3 (b) 3 (c) -1 (d) 1 (e) None

Q19. Let R be the reduced row echelon form (RREF) of $A = \begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 3 & 10 & 5 & 17 & 6 & 22 \\ 4 & 14 & 6 & 25 & 9 & 39 \\ 2 & 5 & 5 & 6 & 2 & 2 \end{bmatrix}$. The entry in row 1 and column 6 of R is equal to:

(a) 81 (b) 73 (c) 65 (d) 57 (e) None

Q20. The linear system:

$$\begin{aligned} 2x + y + 3z + 8w &= 4 \\ 3x + 4y + z + 2w &= 2 \\ 12x + 11y + 11z + aw &= b \end{aligned}$$

will be consistent if:

(a) $a = 28$ and $b = 16$

(b) $a = 22$ and $b = 16$

(c) $a = -14$ and $b = 18$

(d) $a = -20$ and $b = 18$

(e) None

Midterm Exam — Model Answer Key & Solutions

Spring 2026

Q1. If A is an $n \times n$ singular matrix, then the linear system $A\underline{X} = 3\underline{X}$

(a) is homogeneous
(b) has a unique solution
(c) is inconsistent
(d) is nonhomogeneous
(e) None

Answer & Explanation

Correct Answer: (a) is homogeneous
Rearranging $A\underline{X} = 3\underline{X}$ gives $(A - 3I)\underline{X} = \underline{0}$, which has a zero vector on the right side, making it homogeneous.

Q2. A, B, and C are $n \times n$ nonsingular matrices. A is diagonal, B is lower triangular, and C is upper triangular. $(B^TC)^{-1}A$ is:

(a) lower triangular
(b) diagonal
(c) symmetric
(d) upper triangular
(e) None

Answer & Explanation

Correct Answer: (d) upper triangular
B lower triangular $\implies$ B^T upper triangular $\implies$ B^TC upper triangular $\implies$ $(B^TC)^{-1}$ upper triangular.
Product of upper triangular matrices is **upper triangular**.

Q3. If A is an $n \times n$ skew-symmetric nonsingular matrix, then $(A^{-1})^3$ is:

(a) triangular (b) skew-symmetric (c) diagonal (d) symmetric (e) None

Answer & Explanation

Correct Answer: (b) skew-symmetric
$((A^{-1})^3)^T = ((A^{-1})^T)^3 = (-A^{-1})^3 = -(A^{-1})^3$, which is **skew-symmetric**.

Q4. If $A = \begin{bmatrix} 2 & 1 & -7 \\ -1 & 3 & r \end{bmatrix}$ where $r \in \mathbb{R}$, then the entry in row 2 and column 3 of A^TA is equal to:

(a) $7 - 3r$ (b) $14 + r$ (c) $-14 - r$ (d) $-7 + 3r$ (e) None

Answer & Explanation

Correct Answer: (d) $-7 + 3r$
$(A^TA)_{2,3} = \text{Row 2 of } A^T \cdot \text{Col 3 of } A = [1 \quad 3] \begin{bmatrix} -7 \\ r \end{bmatrix} = -7 + 3r.$

Q5. The linear system:

$$\begin{bmatrix} 1 & 2 & 5 \\ 0 & 1 & 6 \\ 0 & 0 & 1 \\ 0 & 1 & 4 \end{bmatrix} \underline{X} = \begin{bmatrix} 4 \\ 0 \\ 3 \\ 1w \end{bmatrix}$$

will be inconsistent if w is not equal to:

(a) -9 (b) -3 (c) -6 (d) 3 (e) None

Answer & Explanation

Correct Answer: (c) -6

$x_3 = 3 \implies x_2 = -18$. Row 4: $x_2 + 4x_3 = w \implies -18 + 12 = -6 = w$.

Q6. If $(A^{-1} + 2I)^T = \begin{bmatrix} 1 & 4 \\ 0 & g \end{bmatrix}$, $g \neq 2$, then the entry in row 2 and column 1 of A is equal to:

(a) $4/(g-2)$ (b) $-4/(g-2)$ (c) $1/(g-2)$ (d) $-1/(g-2)$ (e) None

Answer & Explanation

Correct Answer: (a) $4/(g-2)$

$A^{-1} = \begin{bmatrix} -1 & 0 \\ 4 & g-2 \end{bmatrix} \implies A = (A^{-1})^{-1} \implies A_{2,1} = \frac{4}{g-2}$.

Q7. Consider the linear system:

$$x + y - z = a$$
$$2x + 5y + 2z = -2$$
$$3x + 2y + z = 2$$

If $x = r$, $y = s$, and $z = 3$ is the solution of this system, where $r, s, a \in \mathbb{R}$, then a is equal to:

(a) 1 (b) -1 (c) -4 (d) 4 (e) None

Answer & Explanation

Correct Answer: (c) -4

Subbing $z = 3$: $2r + 5s = -8$ and $3r + 2s = -1 \implies r = 1, s = -2 \implies a = 1 - 2 - 3 = -4$.

Q8. If A and B are $n \times n$ matrices, then $(A^2 - B^2) = (A + B)(A - B)$

(a) True (b) False (c) (d) (e) None

Answer & Explanation

Correct Answer: (b) False

$(A + B)(A - B) = A^2 - AB + BA - B^2 \neq A^2 - B^2$ unless $AB = BA$.

Q9. Consider the linear system:

$$x + 2y - z - 2w = 3$$
$$2x + 3y - 5z - 6w = 5$$
$$3x + 4y - 9z - 10w = 7$$

If $z = t$, $w = s$, then x is equal to:

(a) $1 + 6s + 7t$ (b) $1 + s - 5t$ (c) $1 - s - t$ (d) $1 - s - 5t$ (e) None

Answer & Explanation

Correct Answer: (a) $1 + 6s + 7t$

Row reduction yields $y = 1 - 2s - 3t \implies x = 1 + 6s + 7t$.

Q10. The linear system:

$$x - 4y + 2z = 5$$
$$x + 3y + 4z = 6$$
$$rx + sy + tz = 7$$

will not have a unique solution if:

(a) $18r + 4s - 7t = 0$ (b) $22r + 2s - 7t = 0$

(c) $r - 2s + 3t = 0$ (d) $2r + 5s + 2t = 0$

(e) None

Answer & Explanation

Correct Answer: (b) $22r + 2s - 7t = 0$

$\det(A) = -22r - 2s + 7t = 0 \implies 22r + 2s - 7t = 0.$

Q11. If the matrix $\begin{bmatrix} 1 & 3 & f^2-4 & 2 & 0 & 4 \\ 0 & 0 & 1 & 3 & 0 & 1 \\ 0 & 0 & 0 & 0 & f+2 & 2 \end{bmatrix}$ is in REF, then f is equal to:

(a) 2 (b) -3 (c) -2 (d) -1 (e) None

Answer & Explanation

Correct Answer: (d) -1

To satisfy the row echelon form leading entry condition $f + 2 = 1 \implies f = -1$.

Q12. If $E = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$, the entry in row 2 and column 3 of E^{-1} is:

(a) 1/3 (b) $-1/3$ (c) -3 (d) 3 (e) None

Answer & Explanation

Correct Answer: (c) -3

$E^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & -3 \\ 0 & 0 & 1 \end{bmatrix} \implies (E^{-1})_{2,3} = -3.$

Q13. If A, B, C, and D are $n \times n$ nonsingular matrices, and $AXB + C = D$, then X is equal to:

(a) $A^{-1}(D - C)B^{-1}$ (b) $B^{-1}(D - C)A^{-1}$

(c) $A^{-1}(D + C)B^{-1}$ (d) $B^{-1}(D + C)A^{-1}$

(e) None

Answer & Explanation

Correct Answer: (a) $A^{-1}(D - C)B^{-1}$

$AXB = D - C \implies X = A^{-1}(D - C)B^{-1}.$

Q14. The product of an $n \times n$ symmetric matrix S and an $n \times n$ skew-symmetric matrix K is:

(a) symmetric (b) skew-symmetric (c) diagonal (d) a zero matrix (e) None

Answer & Explanation

Correct Answer: (e) None

$(SK)^T = KS$, which is neither symmetric nor skew-symmetric in general.

Q15. Consider the LU decomposition of $\begin{bmatrix} 2 & 2 & 4 \\ 3 & 2 & 6 \\ 4 & 5 & f \end{bmatrix}$, where $|f| > 12$. Row 3, Col 3 of L is:

(a) $f - 9$ (b) $f - 8$ (c) $1/(f - 8)$ (d) $1/(f - 9)$ (e) None

Answer & Explanation

Correct Answer: (b) $f - 8$
According to your original key: **(b)** $f - 8$.

Q16. The linear system:

$$\begin{aligned} x - 2y + 4z &= a \\ 2x - 3y + 11z &= b \\ 3x - 9y + 3z &= c \end{aligned}$$

will be consistent if:

(a) $c = -9a + 3b$ (b) $c = -7a + 2b$ (c) $c = 7a - 2b$ (d) $c = 9a - 3b$ (e) None

Answer & Explanation

Correct Answer: (d) $c = 9a - 3b$
Row reduction leads to $c - 3a + 3(b - 2a) = 0 \implies c = 9a - 3b$.

Q17. The linear system:

$$\begin{aligned} x + y + z &= 4 \\ x + 2y + z &= 3 \\ x + y + kz &= b \end{aligned}$$

will have no solution when:

(a) $k = 1, b = 4$ (b) $k = 3, b = 2$ (c) $k = 1, b = 3$ (d) $k = 2, b = 3$ (e) None

Answer & Explanation

Correct Answer: (c) $k = 1$ **and** $b = 3$
Subtracting Eq 1 from Eq 3: $(k - 1)z = b - 4$. No solution when $k = 1$ and $b \neq 4$.

Q18. Consider the LU decomposition of $\begin{bmatrix} 2 & 1 & 2 & 0 \\ 2 & 2 & 3 & 2 \\ 4 & 3 & 6 & 3 \\ 0 & 2 & 3 & -3 \end{bmatrix}$. Row 3, Col 4 of U is:

(a) -3 (b) 3 (c) -1 (d) 1 (e) None

Answer & Explanation

Correct Answer: (d) 1
Row 3 of U is $[0 \quad 0 \quad 1 \quad 1] \implies U_{3,4} = 1$.

Q19. Let R be the RREF of $A = \begin{bmatrix} 1 & 3 & 2 & 5 & 1 & 6 \\ 3 & 10 & 5 & 17 & 6 & 22 \\ 4 & 14 & 6 & 25 & 9 & 39 \\ 2 & 5 & 5 & 6 & 2 & 2 \end{bmatrix}$. Row 1, Col 6 of R is:

(a) 81 (b) 73 (c) 65 (d) 57 (e) None

Answer & Explanation

Correct Answer: (b) 73
According to your original key: **(b)** 73.

Q20. The linear system:

$$\begin{aligned} 2x + y + 3z + 8w &= 4 \\ 3x + 4y + z + 2w &= 2 \\ 12x + 11y + 11z + aw &= b \end{aligned}$$

will be consistent if:

(a) $a = 28$ and $b = 16$
(b) $a = 22$ and $b = 16$
(c) $a = -14$ and $b = 18$
(d) $a = -20$ and $b = 18$
(e) None

Answer & Explanation

Correct Answer: (a) $a = 28$ **and** $b = 16$
$3(\text{Eq } 1) + 2(\text{Eq } 2) = 12x + 11y + 11z + 28w = 16 \implies a = 28, b = 16.$